Chronic Kidney Disease :

(Skewness and Kurtosis Analysis)

1. Skewness Report

Skewness measures the asymmetry of the data distribution around the mean.

Variable	Skewness Value	Interpretation	Distribution Shape
sg, hemo	−0.17 to −0.33	Fairly Symmetrical	Nearly a bell curve, though slightly leaning left.
age	-0.66	Moderately Negative	A tail on the left; more older patients than younger ones.
bp, bgr, bu	1.60 to 2.63	Highly Positive	Significant right tail. Most values are low, but a few high values pull the mean up.
sc, pot	7.50,11.58	Extreme Positive	Extremely skewed. This indicates major outliers (e.g., a few patients with very high Potassium).
sod	-6.99	Extreme Negative	Severe tail to the left. Most values are high, but a few extremely low values exist.

2. Kurtosis Report

Kurtosis measures the peakness or flatness of the peak of the frequency distribution.

Variable	Kurtosis Value	Interpretation	Peak/Tail Type
sg	−1.2,−1.14	Platykurtic	Flat peak, thin tails. Very uniform distribution.
age, al, hemo	0.05 to −0.47	Mesokurtic	Very close to a Normal Distribution shape.
su, bgr	5.05,4.22	Leptokurtic	Sharp peak with heavy tails; frequent outliers.
sc, sod, pot	79.3,85.5,142.5	Extremely Leptokurtic	"Infinite" tails. These variables have extreme, rare outliers that deviate wildly from the average.

General Observation:

- **The variables sc (Serum Creatinine), sod (Sodium), and pot (Potassium) are the most "unstable."**
  - Their extreme Skewness and Kurtosis suggest that the "1.5 IQR rule" will likely flag a large number of outliers.
  - For example, the pot Max is 47.0 while the Median is only 4.4. This is why the Kurtosis is a massive 142.5.

- **Hemo (Hemoglobin) and Age** are your most "well-behaved" variables. They follow a roughly normal distribution, making them easier to use in standard statistical models without heavy transformation.

- sg (Specific Gravity) has negative kurtosis. This means the data points aren't clustered in the middle; they are spread out across the range (like a plateau rather than a mountain).

<u>Other Observation:</u>

Since **Mean** is being heavily distorted by extreme values. And since variables like *<u>sc and pot</u>* have such extreme skewness, using Median (strategy="median") or Most Frequent is much safer than using Mean.